



PLATING POSTERS

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PLATING POSTERS INC • METAL FINISHING REFERENCE SERIES

PHOSPHORIC ACID ANODIZING PAA

PROCESS VARIANT **PAA**

= **STRUCTURAL BONDING PREP**

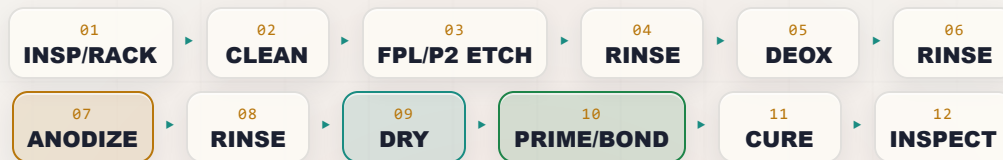
UNSEALED → PRIMED/BONDED

STRUCTURAL ADHESIVE BONDING • AEROSPACE

PHOSPHORIC ACID • THIN BONDING OXIDE • OPEN PORES • LEFT UNSEALED → PRIMED/BONDED

THE COMPLETE TRAINING MANUAL

*A full training course for the brand-new operator — from “wait, the part is the positive electrode?” to a signed certificate. Thin phosphoric-acid anodize on aluminum — the aerospace finish whose whole job is **surface prep for structural adhesive bonding**. We grow a special **very open, whisker-like oxide that adhesives and bond primers sink into and lock onto**. The twist that surprises everyone: **PAA is deliberately left UNSEALED** — sealing would slam shut the very pores the glue needs. Assumes you know nothing. Teaches you everything.*



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • BEGINNER-FRIENDLY STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer/aerospace specifications (e.g., Boeing BAC5555, ASTM D3933, and applicable airframe/customer process specs and bond-primer specs), and current Safety Data Sheets (SDS), and comply with OSHA, EPA, REACH, and local regulations before production use. Phosphoric acid is among the milder anodizing acids — but it still causes acid burns; a dichromate-type FPL pre-etch contains hexavalent chromium (Cr VI, a carcinogen); caustic cleaners burn; hydrogen gas is flammable; and bond primers carry solvent, chromate, and (in some products) isocyanate hazards.

— FIND YOUR WAY

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, your customer's spec (Boeing BAC5555, ASTM D3933, airframe bond-primer and process specs), or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN THE RIGHT PLACE — THE WHOLE POINT HERE IS PREPARING ALUMINUM SO GLUE HOLDS

Welcome. You are about to learn how to run — or at least understand — a **phosphoric acid anodizing line**, the process the trade just calls **PAA**. Whether you started this week or have racked parts for a year and finally want to know *why*, you're in the right place.

Here is the first big idea, and it feels backwards if you've ever seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a thin oxide layer out of the aluminum itself** — with your part hooked up as the **anode** (the *positive* electrode), the exact opposite of plating. That's the single biggest *process* idea in this whole book.

Here is the second big idea, the reason this process exists: **PAA's only job is to prepare an aluminum surface so that adhesives and bond primers grip it and keep gripping it for years**. It builds a **special, extremely open, whisker-like oxide** that glue sinks its roots into — the **gold-standard surface prep for structural adhesive bonding of aircraft aluminum**. Boeing developed and standardized it (BAC5555); the industry leans on it because **PAA bonds survive the moisture and bond-line corrosion that wreck weaker preps**.

Here is the third big idea, the one that trips up every experienced anodizer on a PAA line: **PAA is deliberately left UNSEALED**. On other lines you *seal* the oxide to close the pores; on PAA, sealing would shut the very pores the adhesive needs — so instead **we dry the part and prime or bond it promptly, within a time window**. Burn this in: **decorative/protective anodize → seal; PAA → never seal, prime/bond instead**.



REMEMBER

Do not read "milder acid" as "harmless." Phosphoric acid is gentler than other anodize baths, but the *PAA process chain* still has real hazards: **acid burns, a Cr VI-containing FPL etch (if used), hot caustic cleaner, flammable hydrogen, high-current DC, and bond primers with solvents, chromate, and sometimes isocyanates**. The acid in the tank is the easy part to respect.

The fourth thing: this manual assumes you know nothing about anodizing, chemistry, or electricity — a promise, not an insult. Every term is defined the first time we use it; every step explains *what* and *why*. Don't memorize; *understand*.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, in the order a part actually travels: inspect/rack, clean, **FPL/P2 etch**, rinse, deoxidize, rinse, **PAA anodize**, rinse, **dry**, then **prime or bond — never seal**. The order of a bonding-prep line is not random.



TIP

Read the Fundamentals (Part One) through *before* any station chapter — **especially Chapter 3, on the open oxide and the no-seal rule**. The station chapters assume you grasp the big picture: the part is the anode, oxide grows out of it, the oxide is *open on purpose*, and we *never seal it* — we prime or bond it.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS



TIP

A shortcut or trick of the trade — what a 20-year veteran would lean over and tell you.



REMEMBER

A load-bearing idea worth locking in. If you forget everything else on the page, don't forget these.



HEADS-UP

A safety warning. Read every one — these protect your skin, eyes, lungs, and long-term health.



TECHNICAL STUFF

Deeper chemistry for the curious. Optional — skip it and you can still run the line.



FUN FACT

A bit of trivia to make the science stick (and to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist — what you must *say out loud, unprompted*, before you're cleared — and a bordered **Sign-Off** box your *trainer initials* once you've shown the skill safely. The Teach-Back is your study guide; the Sign-Off is the gate.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **State why PAA exists**: it is a **phosphoric-acid anodize whose only job is surface preparation for structural adhesive bonding** — it builds a thin, extremely open oxide that adhesives and bond primers penetrate and lock into, producing aircraft bonds that resist humidity and bond-line corrosion. Standardized by **Boeing (BAC5555)** and described in **ASTM D3933**.
3. **Describe what the PAA coating does** — a **very thin oxide (commonly a few hundred nanometers up to ~1 µm)** with a **distinctive porous, whisker-like surface** and a thin **phosphate-rich top layer** that together give exceptional, durable adhesion — *not* corrosion protection, *not* wear, *not* decoration.
4. **Explain the no-seal rule** — why PAA is **deliberately left unsealed** (sealing closes the pores the glue needs) and is instead **dried and then primed/bonded promptly within a time window**. This is the single biggest contrast with decorative/protective anodize.
5. **Explain the open porous oxide** — why anodic oxide has pores, why PAA's pores are unusually **open and deep (“whiskery”)**, and why that microscopic geometry is what makes a *mechanically interlocked, durable* bond.
6. **Explain dimensional growth** — the coating grows roughly half in/half out, but because PAA is *extremely thin*, the change is tiny — which is exactly right for tight-fit, fatigue-critical bonded structure.
7. **Place PAA in the family** — Type I (chromic, Cr VI), Type IC / **BSAA** (chrome-free chromic replacement), Type II (sulfuric), Type III (hardcoat), **TSA** (thin sulfuric), and **PAA (the bonding pretreatment)** — and recognize that PAA occupies the **adhesive-bonding niche** the others don't fill.
8. **Explain why the pre-etch matters** — the role of the **FPL etch (Forest Products Laboratory; sulfuric–dichromate)** or the chromate-free **P2 etch**, and why a clean, properly etched/deoxidized surface is mandatory *before* PAA.
9. **Name the stations of the line in order** — ending in **dry** → **prime/bond, never seal** — and explain *why* the order can't be rearranged.
10. **Recognize the main hazards** — phosphoric acid (milder but still corrosive), the **Cr VI in a dichromate FPL etch**, hot caustic cleaner, flammable hydrogen, DC electrical/arcing, and **bond-primer solvents/chromate/isocyanates** — and know the correct PPE, emergency response, and waste handling.

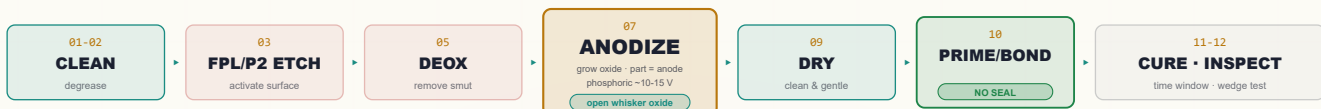


REMEMBER

You are not expected to hit all of these on day one. Anodizing is a craft. But the *safety* objectives (#10) are not optional and not gradual — you must have those before you work the line at all.

• THE LINE AT A GLANCE

Don't memorize yet — feel the rhythm: **Inspect/Rack** → **Clean** → **FPL/P2 Etch** → **Rinse** → **Deox** → **Rinse** → **PAA ANODIZE** → **Rinse** → **DRY** → **PRIME/BOND** → **Cure** → **Inspect**. (Notice what's missing versus a decorative line: *no Dye and no Seal*. The line ends by *gluing, not by closing pores*.)



REMEMBER

Notice the pattern: **a rinse follows almost every working tank**. Rinses aren't breaks — they're checkpoints that keep one chemistry from poisoning the next *and* keep the bonding surface clean. On a PAA line, the cleanliness of the surface you hand to the primer is everything; a careless rinse or a fingerprint can quietly wreck a structural bond.



FUN FACT

The whisker-like oxide PAA grows is so distinctive that researchers identify it by electron microscope on sight — a microscopic forest of tiny columns and pores. That “forest” is precisely what the adhesive grows roots into. PAA doesn't just give the glue a *clean* surface; it gives it a *3-D scaffold*.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a thin oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. Now let's unpack it, because it's genuinely different from plating and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you hook your part up as the **cathode** (the *negative* electrode) and *deposit a different metal* onto it from a bath. The coating is a *separate metal sitting on top of* your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word “anodize” comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate.



REMEMBER — THE SINGLE BIGGEST IDEA IN THIS BOOK

In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself**. If you remember nothing else from Chapter 1, remember this. Everything else follows from it.

HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

Picture a tank full of liquid called the **electrolyte** (or just “the bath”). For PAA, that liquid is **dilute phosphoric acid** — phosphoric acid mixed with water. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction “DC” current anodizing needs):

- **The anode** — your **aluminum part**, connected to the **positive** terminal. “Anode = the part, the thing growing oxide.”
- **The cathode** — bars or plates (commonly **stainless steel or lead**) hanging in the same tank, connected to the **negative** terminal. The cathode just completes the circuit and is where **hydrogen gas** bubbles off.



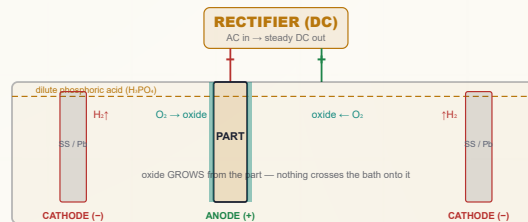
FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it. On a PAA part the “coating” is so thin you often can't see it as a color at all — but it changes how that surface bonds more than a thick coat of anything else could.

When you turn on the rectifier:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al_2O_3)**.
3. The phosphoric acid gently dissolves the oxide as it forms — and with phosphoric acid this tug-of-war produces a **very open, deeply porous, whisker-like structure** (Chapter 3). Phosphate ions are also taken up into the oxide, leaving a thin **phosphate-rich top layer**.
4. Over at the cathode, **hydrogen gas** bubbles off (a safety item — flammable gas — more later).

The net result: the surface of your part *converts* into a controlled, extremely open porous oxide — purpose-built not to protect or color the part, but to be **the best glue-able surface aluminum can have**.



Compare to a plating cell: the labels are reversed — here the **part is the ANODE (+)** and the counter-electrodes are the cathodes.



TECHNICAL STUFF — THE TWO REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2\text{Al} + 3\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 6\text{H}^+ + 6\text{e}^-$. At the cathode, those electrons reduce hydrogen ions: $6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2\uparrow$. So **oxygen builds the coating at your part; hydrogen gas leaves at the counter-electrode**. No aluminum travels through the bath onto something else — it stays put and turns to oxide in place. In phosphoric acid, some **phosphate is incorporated** into the outermost oxide, part of why PAA surfaces resist hydration (the enemy of durable bonds).



TECHNICAL STUFF — WHY PAA IS “VOLTAGE-CONTROLLED,” NOT “AMP-HOUR-CONTROLLED”

On a thick decorative line you push a set current density for a set time and the oxide keeps growing. PAA is different: you **ramp to a set voltage (~10–15 V) and hold it**. As the thin oxide forms, it resists current flow, so the **current naturally tapers off** and the oxide reaches a thin, stable, self-limited thickness. You're not trying to grow a thick coating — you're trying to grow the *right open structure* and stop. *Thin is the goal, not a limitation.*

WHAT IS “PAA”?

THE AEROSPACE BONDING PRETREATMENT — BUILT TO MAKE ALUMINUM GLUE JOINTS LAST THE LIFE OF THE AIRFRAME

PAA (phosphoric acid anodizing) is a phosphoric-acid anodize used as surface preparation for structural adhesive bonding. It grows a very thin (commonly a few hundred nanometers up to about 1 µm) aluminum-oxide layer with a **uniquely open, porous, whisker-like surface** and a thin **phosphate-rich skin**. It was **developed and standardized by Boeing (process spec BAC5555)** and is described in **ASTM D3933** (“phosphoric acid anodizing for structural adhesive bonding”). When an aircraft has bonded aluminum structure — honeycomb sandwich panels, bonded skins and doublers, laminated repairs — PAA is very often the surface prep underneath that bond.

• WHY AEROSPACE BONDS WITH IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Durable bonds	The open whisker oxide gives adhesives a 3-D mechanical interlock — bonds that survive heat and humidity far better than bonds made on a simply cleaned or etched surface.
Resists bond-line corrosion	The phosphate-rich, hydration-resistant oxide slows the moisture attack that creeps along weak bond lines and peels joints apart over years.
Very thin / low distortion	A few hundred nm of oxide barely changes dimensions — ideal for tight-fitting, fatigue-critical bonded assemblies.
Compatible with bond primers	The open surface is exactly what corrosion-inhibiting adhesive primers (e.g., BR 127-type primers) are designed to wet and anchor into.
Proven & specified	Decades of aerospace service history; called out by Boeing BAC5555, ASTM D3933, and many airframe/customer specs.

WHERE YOU’LL SEE IT

Bonded aircraft structure: **honeycomb sandwich panels, fuselage/wing skin doublers, bonded stringers and fittings, laminated and repair bonds**. PAA is a *bonding-shop* and *aerospace-repair* process far more than a general job-shop finish.



REMEMBER

PAA’s whole identity is in one phrase: **surface prep for structural bonding**. It is *not* there to protect the part, color it, or harden it. If someone asks “how thick is the PAA coating?” the honest answer is “thin on purpose — the point isn’t thickness, it’s an open, bondable surface that’s left unsealed and primed promptly.”



TIP

When a print or traveler calls out PAA / BAC5555 / ASTM D3933, expect the part to go to **bonding or priming next** — not to a seal tank, not to a dye tank. Your job is to hand off the cleanest, most undisturbed open oxide you can, on time.



FUN FACT

Before PAA became standard, many aluminum aircraft bonds were prepped only with the FPL acid etch. Bonds were strong *when new* but could lose strength in humid service. The move to PAA was largely about **durability** — keeping bonds strong for *decades*, not just on day one.

THE OPEN OXIDE

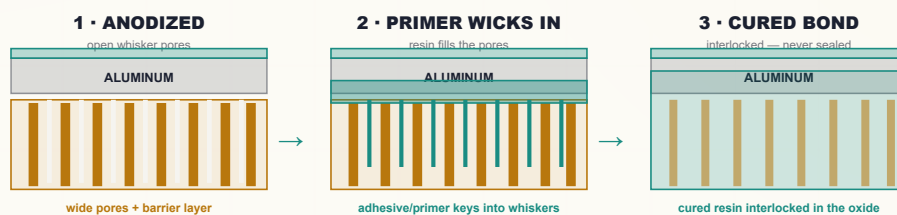
THE STRANGEST, MOST IMPORTANT IDEA ON A PAA LINE — AND THE REASON WE NEVER, EVER SEAL IT

This is the chapter that makes PAA *PAA*. Slow down here. On every other anodize line you learn “anodize → (dye) → seal.” On a PAA line you must *un-learn the seal*.

THE OXIDE GROWS WITH MICROSCOPIC HOLES IN IT

Remember the tug-of-war from Chapter 1: oxide forms at the surface while the acid gently dissolves it. The result is not a smooth glassy sheet — it's a layer made of **billions of tiny columns with a microscopic pore down the center of each**, sitting on a thin, dense **barrier layer** against the metal. *Every* anodize has this. What makes **phosphoric acid** special is that it produces an **unusually open, deep, large-pore “whisker” structure** — the most penetrable oxide of any common anodize.

So the finished PAA oxide has: a thin dense **barrier layer** at the bottom, a thin **porous/whisker layer** above it with **wide-open pores**, and a thin **phosphate-rich skin** on top. Those wide-open pores and whiskers are the whole point — because **anything you can get into the pores locks into the coating**. On a decorative line that “anything” is dye. On a PAA line that “anything” is **adhesive and bond primer**.



On PAA the pores are filled by **ADHESIVE**, not closed by sealing.

WHAT REPLACES “DYE”: THE ADHESIVE/PRIMER WICKS IN

Right after anodizing and drying, the pores are open and thirsty. When the **bond primer or adhesive** touches that surface, it **wicks down into the whiskers and pores** and cures there — so the cured polymer is *mechanically interlocked* with the oxide, not just stuck to a smooth face. That interlock is why PAA bonds are so strong and so durable.



REMEMBER — THE HEART OF PAA: DO NOT SEAL

On a decorative/protective line, the *last* chemical step is **sealing** — closing the pores with near-boiling water or nickel salts. On a **PAA** line we **never seal**. Sealing would swell the oxide and **slam the pores shut** — destroying the open structure the adhesive needs. After PAA we **dry the part and prime/bond it**. Sealing a PAA part ruins it as surely as forgetting to dye a colored part.

WHAT DOES NOT HAPPEN: SEALING

Here is the rule that separates PAA from almost everything else you may have learned:



HEADS-UP — “ANODIZE → SEAL” IS MUSCLE MEMORY; BREAK IT HERE

Anyone trained on Type II or BSAA will instinctively reach for a seal tank. On a PAA line that instinct is a *scrap-the-lot* mistake. There may not even *be* a seal tank — and if there’s a hot-water rinse, it must be controlled so it doesn’t accidentally start sealing the oxide. **When in doubt, do not apply heat that could seal the surface, and ask your supervisor.**



TECHNICAL STUFF — WHY “NO SEAL” AND “STILL DURABLE”

Sealing exists on other lines to close pores so the bare anodize resists corrosion and locks in dye. PAA achieves *bond durability* a different way: the **phosphate-rich oxide resists hydration** (water attack) and the **open structure is filled by cured primer/adhesive**, which itself protects the interface. So PAA gets long-term durability **from the chemistry and the bond**, not from sealing. Close the pores and you’d lose the interlock *and* gain nothing PAA needs.



TECHNICAL STUFF — THE PHOSPHATE TOP LAYER

Phosphoric acid leaves a thin layer of **aluminum phosphate** at the very outer surface. This layer is more resistant to converting to hydrated oxide (boehmite) in humid air than a plain sulfuric oxide would be — one reason PAA surfaces tolerate a (limited) wait before priming and give such hydration-stable bonds. It is still not a license to leave parts exposed; the time window (Station 11 / Part Three) is real.

THE ONE-LINE RULE TO CARRY OFF THIS PAGE

Decorative / protective anodize → seal. PAA → never seal; dry, then prime/bond promptly. Everything in Part Two’s back half (Stations 08–11) is just the careful execution of that one rule: keep the open oxide clean, dry it gently, and fill it with primer/adhesive before it ages.



REMEMBER

If you take one rule from this entire manual: **PAA is primed/bonded, not sealed.** A sealed PAA part has closed pores, no mechanical interlock, and a weak, undurable bond — it looks finished but it is scrap.



FUN FACT

On a PAA part you can’t “see” the prep the way you see a bright blue dyed carabiner. The magic is invisible to the eye and visible only to an electron microscope — and to the test engineer who pulls a bonded coupon apart months later and finds it still holding.

CHAPTER 4

DIMENSIONAL GROWTH

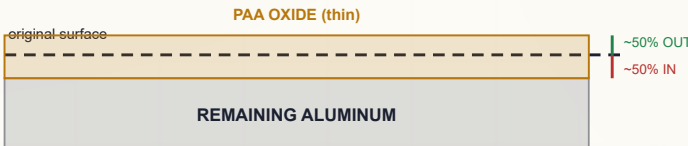
THE COATING MAKES THE PART BIGGER — BUT ON PAA, GLORIOUSLY LITTLE

Because we're converting aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the coating grows in **two directions at once**:

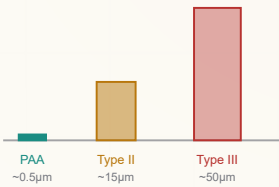
- About **half** grows *into* the original surface (consuming aluminum).
- About **half** grows *out* from the original surface (the part gets bigger).

That's the same "Pilling–Bedworth" physics as any anodize. **But PAA is extremely thin** — a few hundred nanometers up to about a micron — so the dimensional change is **tiny**, far smaller than Type II (~5–25 µm) or Type III hardcoat (~25–100+ µm). For bonded, fatigue-critical, close-tolerance aerospace structure, **"almost no dimensional change" is exactly what you want**.

PAA: a few hundred nm → change essentially negligible



Thickness scale



REMEMBER — THE 50% RULE, BUT WITH A TINY NUMBER

Anodize grows **roughly half in, half out**. On PAA the *total* is so small that on most parts you treat the dimensional change as negligible — but you must still **mask** any surface that must not be anodized (electrical grounds, certain fits) because the oxide is **electrically insulating** even though it's thin.



TIP

Even though PAA growth is tiny, **don't anodize surfaces that need to conduct electricity or that aren't part of the bond**. A thin insulating oxide still insulates. Mask grounds and contact points per the print.



HEADS-UP — MASKING AND THE BOND AREA

The opposite concern matters more on PAA: make sure the **surfaces that will be bonded are not masked, not handled, and not contaminated**. A masked or fingerprinted faying surface is a weak bond waiting to fail.



TECHNICAL STUFF — THE PILLING–BEDWORTH IDEA

Aluminum oxide occupies more volume than the aluminum it formed from (Pilling–Bedworth ratio > 1). That volume expansion is *why* any anodize grows outward — and it's also *why sealing works* on other lines (the oxide swells shut). On PAA we **want** that expansion to *not* go to completion, one more reason we keep the oxide thin and never seal.



FUN FACT

On Type III hardcoat, drawings often call out separate pre- and post-anodize dimensions because the growth is big. On PAA the growth is so small that the **bond** — not the dimension — is what the engineering drawing cares about.

CHAPTER 5

THE TYPE FAMILY

SAME IDEA, MANY FLAVORS — AND PAA OWNS THE BONDING CORNER NONE OF THE OTHERS FILL

All of these grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, thickness, and purpose**.

	TYPE I — CHROMIC	TYPE IC / BSAA	TYPE II — SULFURIC	TYPE III — HARDCOAT	PAA (THIS MANUAL)
Electrolyte	Chromic acid (Cr VI)	Boric-sulfuric	Sulfuric (~165–200 g/L)	Sulfuric (cold/conc.)	Phosphoric (~10–12%)
Typical thickness	~0.5–7.5 μm	~2–7 μm	~5–25 μm	~25–100+ μm	~0.1–1 μm
Pore / oxide	Fine	Fine, thin	Porous, dyeable	Dense, hard	Very open, whisker-like
Sealed?	Usually	Often (or unsealed for paint)	Yes (dye + seal)	Often	Never — left unsealed
End step	Seal	Seal / paint	Dye + seal	Seal	Prime / bond
Main purpose	Aerospace corrosion / paint base	Chrome-free chromic replacement	Decorative + protective workhorse	Maximum hardness & wear	Structural adhesive bonding prep

**REMEMBER**

Type I / BSAA = thin corrosion/paint-base pretreatments. Type II / III = thick functional/decorative coatings. PAA = a thin BONDING pretreatment — open pores for adhesive, left unsealed, primed not painted. PAA is the only member of the family whose job is to be *glued*.

**TIP — THE TWO COUSINS YOU’LL HEAR ABOUT**

Sitting near PAA are **TSA** (thin-film sulfuric anodize) and **BSAA** (boric-sulfuric) — both chrome-free aerospace pretreatments aimed mainly at **corrosion protection and paint base**. They *can* serve as bond preps in some specs, but **PAA remains the gold standard for the most durable structural bonds**. Each has its own manual in this series; they all stand on the anodizing fundamentals you’re learning now.

**TECHNICAL STUFF — WHY PHOSPHORIC FOR BONDING**

Each acid grows a different pore geometry. Sulfuric grows fine pores; chromic grows a thin, gentle oxide; **phosphoric grows the most open, deep, large-cell oxide with a hydration-resistant phosphate skin**. That open geometry plus hydration resistance is the recipe for adhesive penetration *and* long-term bond durability — which is exactly why Boeing built the PAA process (BAC5555) around phosphoric acid for bonding rather than reusing a sulfuric bath.

**FUN FACT**

Chromic acid once did several jobs at once — corrosion pretreatment, paint base, *and* bonding prep. Replacing it without Cr VI took a small *family*: BSAA and TSA for general paint-base duties, and **PAA for the most demanding adhesive bonding**.

CHAPTER 6

ALUMINUM ALLOYS

ANODIZING ONLY WORKS ON ALUMINUM — AND THE AEROSPACE BONDING ALLOYS HAVE THEIR OWN QUIRKS

Anodizing only works on **aluminum** and a few other “valve metals.” On a PAA line the parts are almost always **aerospace aluminum**, which means you’ll see a lot of the **copper-rich 2xxx (especially 2024)** and **zinc-rich 7xxx (especially 7075)** structural alloys, plus **clad sheet** (a thin pure-aluminum layer roll-bonded onto a strong core for corrosion resistance).

THE AEROSPACE ALLOYS YOU’LL ACTUALLY RUN

- **2024 (Al-Cu)**: the classic bonded-structure alloy. **Copper** in the alloy leaves heavy **smut** after etching and can dull the oxide — so deox/desmut discipline matters.
- **7075 (Al-Zn)**: high-strength structure; also forms smut and is sensitive to over-aggressive etching.
- **Clad (Alclad) sheet**: the pure-Al cladding anodizes very cleanly — but it is **thin**, so over-etching can cut through the clad layer and expose the copper-rich core, ruining both corrosion protection and bond quality.
- **6xxx / 5xxx**: anodize cleanly; less common on primary bonded structure but seen in fittings and repairs.

ALLOY FAMILY	MAIN ALLOYING ELEMENT	PAA / BONDING BEHAVIOR
2xxx (e.g., 2024)	Copper	Heavy smut; control etch & deox; still a top bonding alloy
7xxx (e.g., 7075)	Zinc (+ Cu/Mg)	Smut-prone; avoid over-etch
Clad (Alclad)	Pure-Al skin over core	Bonds beautifully — but don’t etch through the thin clad
5xxx / 6xxx	Mg / Mg-Si	Anodize cleanly; good bonding
Cast (high-Si)	Silicon	Gray/mottled; rarely PAA’d for primary bonds



REMEMBER

On bonding alloys, two alloy facts dominate: **copper makes smut** (so deox/desmut is non-negotiable) and **clad layers are thin** (so over-etching is dangerous). Both come back to *controlling the etch*, not just running it.



TIP

Always confirm the **alloy and whether it’s clad** before you set etch time. The right etch for bare 2024 can cut too deep into thin Alclad. When unsure, check the traveler and ask — a bonded primary-structure part is not the place to guess.



FUN FACT

“Alclad” is a corrosion trick from the 1920s: a thin skin of nearly pure aluminum is roll-bonded onto strong copper-bearing alloy so the skin corrodes sacrificially instead of the structure. It’s wonderful for corrosion — and a reminder to your etch timer that there isn’t much skin to spare.

CHAPTER 7

HOW THE LINE WORKS

WALKING THE LINE — EVERY TANK HAS A JOB, AND THE ORDER IS THE RECIPE (AND IT ENDS IN GLUE, NOT SEAL)

A PAA / bonding-prep line is a sequence of tanks and steps. Here's the whole journey and *why each step has to be where it is*.

1. **Inspect / Rack** — Check the part, confirm alloy/spec, and rack it for firm **electrical contact** (the part *is* the anode) and so the **bond surfaces stay clean and accessible**.
2. **Clean / Degrease** — Remove oils and shop soil. Oxide won't grow evenly through grease, and grease is the enemy of bonding.
3. **FPL / P2 Etch** — A controlled acid etch that **deoxidizes and chemically activates** the surface and starts building the right microstructure. The classic **FPL etch is sulfuric–sodium-dichromate (contains Cr VI)**; the chromate-free **P2 etch (ferric-sulfate / sulfuric)** is the modern substitute.
4. **Rinse** — Wash the etch off before anodizing; protect the anodize bath and the surface.
5. **Deoxidize / Desmut (alloy-dependent)** — On copper-rich alloys, remove residual **smut** so the surface is uniform and bondable. (On some flows the FPL/P2 etch does this job; follow your spec.)
6. **Rinse** — Clean handoff before the main tank.
7. **PAA ANODIZE** — The main event. Part = anode, **dilute phosphoric acid, ~10–15 V**, room temperature, ~20–25 min. Grow the thin, open, whisker oxide.
8. **Rinse** — Gently rinse phosphoric acid out of the fresh open pores **without disturbing or contaminating them**.
9. **Dry** — A **clean, gentle dry** (this *replaces* the “dye” step of a decorative line). The surface must be dry and uncontaminated for priming/bonding.
10. **Prime / Bond — NO SEAL** — Apply the **corrosion-inhibiting bond primer** and/or the **adhesive**, promptly. (*This replaces the “seal” step. We close nothing — we fill the pores with primer/adhesive.*)
11. **Cure / Assemble** — Cure the primer/adhesive and lay up the bonded assembly, **within the time window** before the surface degrades.
12. **Inspect** — Verify surface prep and **bond quality** (including the wedge test, ASTM D3762, on process-control coupons).



REMEMBER — WHY THE ORDER CAN'T MOVE

You can't etch before you clean (grease blocks the etch). You can't anodize before you deox (smut ruins the oxide). You can't bond a wet or dirty surface (weak bond). And you **never seal** — you **prime/bond** instead, **on time**. The line *is* the recipe.



HEADS-UP — THE LINE IS A TOUR OF HAZARDS

As you walk the line you pass **hot caustic cleaner**, a **Cr VI–bearing FPL etch (if used)**, **phosphoric acid**, **flammable hydrogen** (cathode), **high-current DC**, and **bond-primer solvents/chromate/isocyanates**. Each station chapter calls out its own. Respect each step for what *it* is.



TIP

When you're learning, walk the empty line with your trainer and name each tank's *job* and *main hazard* out loud — and at the end, say the two PAA-specific rules: **“never seal”** and **“prime/bond within the time window.”** If you can do that, the rest is detail.

CHAPTER 8

PPE — YOUR DAILY ARMOR

WHAT YOU WEAR, AND WHY EACH PIECE IS THERE

A PAA / bonding line works with phosphoric acid, an acid (possibly chromate) etch, hot caustic cleaner, high-current DC, hydrogen, and **bond primers with solvents and chromate**. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

HAZARD ON THE LINE	PPE THAT PROTECTS YOU
Phosphoric acid (anodize) — burns, splash	Sealed chemical splash goggles + face shield , chemical-resistant gloves, apron/suit, chemical-resistant boots
FPL etch (sulfuric–dichromate, Cr VI) or P2 etch — burns + carcinogen	Same acid PPE; for dichromate FPL, chromate-rated gloves, strict no-skin-contact, respiratory protection per program
Hot caustic cleaner — deep burns	Same acid/caustic PPE; caustic burns are deep and slow to feel
Acid mist / chromate mist	Local exhaust at tanks; respiratory protection per your program (especially over a dichromate etch)
DC electrical / arcing at contacts	Dry hands, dry gloves, no jewelry, follow LOTO
Bond-primer solvents, chromate pigments, (some) isocyanates	Respirator per program (often supplied-air for spray priming), solvent-rated gloves, eye protection, ventilated spray area
Slips on wet floors	Non-slip chemical-resistant boots

★

REMEMBER

On a PAA line your **eyes, skin, and lungs** are all exposed. The *acid* is milder than other anodize baths — but the **chromate (FPL etch and many primers)** and the **primer solvents/isocyanates** make **respiratory protection and skin protection just as important here as on any line**. Goggles + face shield at the wet tanks are not optional; respiratory protection in the etch/prime areas is governed by your program.

⚠

HEADS-UP — GOGGLES FOR CHEMICALS, NOT JUST SAFETY GLASSES

Safety glasses stop flying chips; they do **not** stop an acid, chromate, or solvent splash from the side. At every chemical tank and in the prime area, sealed splash goggles (and a face shield when charging, dumping, or handling parts) are required.

⚠

HEADS-UP — NEVER EAT, DRINK, SMOKE, VAPE, OR APPLY COSMETICS ON THE LINE

Hand-to-mouth contact is a major route for ingesting chemicals — **especially chromate** from a dichromate FPL etch or chromated primer. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

💡

TIP

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. Rinse your gloves *before* you take them off. In the prime area, treat your respirator as the most important item, not an afterthought.

CHAPTER 9

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read.

SITUATION	WHAT TO DO — IMMEDIATELY
Acid or caustic on skin	Flush immediately with water for at least 15 minutes . Remove contaminated clothing while flushing. Get medical attention — caustic burns can be deep even when they don't hurt at first.
Acid or caustic in eyes	Go straight to the eyewash , hold eyes open, flush at least 15 minutes , get medical help. Don't stop early.
Chromate (FPL etch / primer) on skin or in eyes	Flush 15 minutes; chromates are carcinogenic and sensitizing — get medical attention and report exposure; follow the SDS.
Large splash / drench	Use the safety shower — full 15 minutes, clothing off.
Fluoride exposure (some deox/desmut chemistries)	Fluoride burns are deep, delayed, and systemically toxic — flush, then apply the specific first-aid in the SDS (often calcium gluconate gel) and get medical help immediately. Ordinary flushing alone may not be enough.
Solvent / isocyanate primer exposure or fumes	Move to fresh air; for breathing trouble get medical help. Isocyanate sensitization can be permanent — respiratory protection is prevention . Follow the primer SDS.
Electrical / arc at rectifier or contacts	Don't touch — de-energize via rectifier controls / LOTO. Never reach into a tank with current on.
Hydrogen gas	Keep ignition sources (flames, sparks, grinding) away from the cathode/tank area; ensure ventilation is running.

HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP

Always add **ACID** to **WATER**, never **water** to **acid**. (“Do as you *oughta*: add acid to water.”) This applies to **phosphoric acid** and to the **sulfuric** in the **FPL/P2 etch**. Bath make-up/additions are usually authorized-personnel work — but every operator must know this rule.

HEADS-UP — LOCKOUT/TAGOUT (LOTO)

LOTO is mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it so no one can re-energize while someone is working. “I’ll just be a second” is how people get electrocuted.

REMEMBER

Know where the **nearest eyewash, safety shower, spill kit, and (in the prime area) fresh-air route** are at *every* station before you start your shift. In a chemical emergency you won't have time to look. **Eyewash and shower stations must always be unblocked.**

CHAPTER 10

THE SAFETY PHILOSOPHY OF A PAA / BONDING LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER

1. TWO CORROSIVES, MIRROR-IMAGE.

The **cleaner is caustic** (high pH); the **etch and anodize are acid** (low pH). Both burn. Treat high pH and low pH with equal respect — and never let them mix outside a controlled neutralization.

2. THE CARCINOGEN MAY BE IN THE *ETCH*, NOT THE ANODIZE.

Unlike chromic anodizing, PAA's *anodize* bath has no Cr VI — but a classic **FPL etch is sulfuric–dichromate (Cr VI)**, and many **bond primers contain chromate**. On this line the chromium hazard hides in the etch and the primer. Know which chemistry your shop runs; **chromate-free P2 etch and chromate-free primers exist** and may be specified.

3. THE PART IS LIVE.

Because the part is the anode, there is **DC current at the work** and at the contacts. Wet hands + current + acid is a bad combination.

4. HYDROGEN IS FLAMMABLE.

The cathode bubbles off hydrogen. Ventilation on, ignition sources away.

5. THE PRIME AREA IS ITS OWN WORLD.

Solvents, chromate pigments, and (in some primers) isocyanates mean the **prime/bond step has respiratory hazards** the wet line doesn't. Treat it with its own PPE and ventilation discipline.

6. ENGINEERING CONTROLS FIRST, PPE SECOND — AND WHEN IN DOUBT, ASK.

Tank exhaust, spray-booth ventilation, splash guards, and cooling do the heavy lifting; PPE is your last layer, not your first. The unsafe operator isn't the one who asks questions — it's the one who guesses.

• THE HAZARD FAMILIES TO INTERNALIZE

1. **Corrosives (acid AND caustic).** Phosphoric (anodize), sulfuric ± dichromate (etch), caustic (cleaner). *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
2. **Toxics (chromate, fluoride, isocyanate).** Cr VI in a dichromate FPL etch / chromated primer; fluoride in some deox; isocyanate in some primers. *Defense:* respiratory protection, strict no-skin-contact, hygiene, specific first-aid (calcium gluconate for fluoride), and **substitution where specs allow (P2 etch, chrome-free primer)**.
3. **Electrical & gas.** High-current DC; flammable hydrogen. *Defense:* LOTO, dry hands, no ignition sources, ventilation.



REMEMBER

On most anodize lines the headline hazard is the *bath acid*. On a PAA line the headline hazards are just as much the **chromate in the etch/primer and the solvents in the prime area**. Phosphoric acid is the milder part of this process — don't let "milder acid" lull you about the rest of the chain.



REMEMBER

You now have the whole foundation: what anodizing is (and how it's *not* plating), part-is-the-anode/grow-oxide, the open whisker oxide, **the no-seal/prime-instead rule**, tiny dimensional growth, the anodize family and PAA's bonding niche, the bonding alloys, how the line works, what to wear, and the safety mindset. Every station chapter builds on this.

PART TWO — STATION BY STATION
STATION 01 OF 12

INSPECTION & RACKING

“IN ANODIZING, THE PART IS THE WIRE. A LOOSE PART IS A DEAD PART — AND ON A BONDED PART, A DIRTY SURFACE IS A DEAD BOND.”

PURPOSE & THE “WHY”

Inspect the incoming aluminum part, confirm the **alloy/clad and the spec** (PAA per BAC5555 / ASTM D3933, which surfaces get bonded, what’s masked), and **rack it** so it makes firm current-carrying contact, hangs for good drainage and gas escape, and — critically — so the **bond surfaces stay clean and untouched**. Because the part is the **anode**, all the current that grows your oxide flows through the rack contact. A loose contact means **no oxide or a burned spot**. And because this part will be **glued**, the cleanliness and integrity of the faying (bond) surfaces is just as important as the electrical contact.



REMEMBER — TWO NO-GO ZONES

(1) Wherever the rack grips, **no oxide grows** (that spot carries current). Place the contact where the bare mark won’t matter — never on a bond surface. (2) **Never touch, glove-smudge, or mask a faying surface** that’s supposed to bond. On PAA, racking protects *both* the electrical contact *and* the future bond.

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / print	Confirm alloy, clad y/n , PAA spec, bond surfaces, masked areas
Rack material	Titanium (preferred) or aluminum	Ti racks don’t anodize away; Al racks anodize and need stripping
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, no coat, dropped parts
Contact location	Hidden / non-bond area	The contact point won’t anodize (bare mark) — keep it off bond surfaces
Masking	Plugs/tape on no-anodize areas only	Never mask a surface that must bond
Orientation	Gas escape & drainage	Avoid trapped air (un-anodized voids on a bond surface = weak bond)



HEADS-UP

Racks and parts have **sharp edges and pinch points**, handled near chemical tanks. A part that drops from a weak rack becomes a **splash hazard** in an acid tank, not just scrap. Watch your hands.



TIP — HANDLE BY THE RACK, AND KEEP BOND SURFACES PRISTINE

From racking onward, handle parts **by the rack only**. A bare-hand fingerprint on a faying surface puts oils exactly where the adhesive must grip. Clean gloves, clean racks, no shortcuts.

STEP-BY-STEP

1. Read the traveler (alloy, clad, PAA spec, bond surfaces, masked areas). 2. Inspect for scratches/gouges/prior coatings — anodize *reveals* flaws. 3. Plan a hidden contact point off any bond surface. 4. Mask only true no-anodize areas (never a bond surface). 5. Rack firmly — good metal-to-metal contact, correct orientation. 6. Verify no parts touch (shadowing). 7. Handle by the rack from here on.

SIGN-OFF — STATION 01

Teach-Back: Why contact matters more in anodizing (part = anode); why the contact point won’t anodize; why titanium racks are preferred; why orientation matters for gas/drainage; and **why bond surfaces must never be masked, touched, or shadowed**.

“I verified alloy/clad and PAA spec, chose a firm contact in a non-bond area, masked only true no-anodize features, racked for gas escape and drainage with no parts touching and bond surfaces protected, and handled parts by the rack.”

Trainee: _____ Trainer: _____ Date: _____

STATION 02 OF 12

CLEAN / DEGREASE

“ANODIZE OVER OIL AND YOU’VE GROWN A DEFECT. BOND OVER OIL AND YOU’VE BUILT A FAILURE.”

PURPOSE & THE “WHY”

Get the part **chemically clean** — no oil, grease, fingerprints, or shop soil — so the etch and oxide work evenly and the future bond has clean metal to anchor to. Cleaning is usually a **vapor degrease and/or a mild alkaline (non-etching) soak cleaner** — formulated *not* to attack the aluminum (we save controlled metal removal for the etch). The free, foolproof check is the **water-break test**.



TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds. **On a bonding line, treat a water-break failure as a bond-reject — never anodize or bond a part that beads.**

FAIL - water beads up

```
+-----+
| o o o | X
| o o |
+-----+
```

Oil remains -> RE-CLEAN

PASS - water sheets evenly

```
+-----+
|#####| OK
|#####|
+-----+
```

Surface clean -> proceed

PARAMETER	RANGE	WHAT TO WATCH
Type	Vapor degrease and/or mild non-etching alkaline cleaner	Must not attack aluminum
Temperature	~50-70 °C (120-160 °F)	Per cleaner TDS
Time	~3-10 min	Heavy oil longer
Concentration	Per TDS	Titrate regularly
Water-break test	Pass = continuous sheet	The only reliable field check — mandatory on bond parts



HEADS-UP — HOT ALKALINE SOLUTION

Even a “mild” cleaner is hot and alkaline: splashes irritate/burn skin and eyes; steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.



HEADS-UP — SOLVENT VAPOR DEGREASE

If your shop vapor-degreases, the **solvent vapors** are a respiratory and (some solvents) health hazard. Use it per its SDS with proper ventilation; never lean into the vapor zone.

STEP-BY-STEP

1. Check tank temperature/concentration in range. 2. Immerse fully; start timer; confirm agitation. 3. Withdraw slowly, draining back. 4. Run the **water-break test** — sheet → proceed; beading → re-clean. 5. Move promptly to the etch (don’t let a clean part sit and re-contaminate).

SIGN-OFF — STATION 02

Teach-Back: What the cleaner removes (oil/soil) vs. what it must *not* do (attack aluminum — that’s the etch); perform/read a water-break test; why a clean surface is doubly critical on a bonding line; the caustic/solvent hazards and PPE.

“The part passed the water-break test, the cleaner was in range and at temperature, and I wore required PPE.”

Trainee: _____ Trainer: _____ Date: _____

STATION 03 OF 12

FPL / P2 ETCH

“THE STEP THAT ACTIVATES THE METAL FOR BOTH THE ANODIZE AND THE BOND — AND WHERE THE CR VI MAY HIDE.”

PURPOSE & THE “WHY”

Chemically **deoxidize and activate** the aluminum and begin building the right microstructure, using a controlled acid etch, so the PAA anodize grows a uniform open oxide and the surface bonds well. There are two common chemistries:

- **FPL etch (Forest Products Laboratory):** the classic **sulfuric acid + sodium dichromate** etch, run warm (~60–66 °C). It works beautifully — but the **dichromate is hexavalent chromium (Cr VI), a carcinogen**.
- **P2 etch:** a **chromate-free** replacement (a **ferric-sulfate / sulfuric-acid** etch, developed as a drop-in for FPL). Same job, **no Cr VI**. Many shops have switched.

Verify before use: exact FPL and P2 bath make-ups, temperatures, and times vary by spec — confirm against your customer process spec, supplier TDS, and SDS. The Cr VI status of your specific etch is the load-bearing safety fact to confirm.



REMEMBER — THE ETCH SETS UP THE BOND

On a decorative line the etch just gives a satin look. On a **PAA bonding line the etch is bond-critical**: it deoxidizes, removes smut, and conditions the surface so the anodize and the adhesive both work. Get the etch wrong and no amount of careful anodizing or priming saves the bond.

PARAMETER	FPL (CLASSIC)	P2 (CHROME-FREE)
Chemistry	Sulfuric acid + sodium dichromate (Cr VI)	Ferric sulfate + sulfuric acid (no Cr VI)
Temperature	~60–66 °C (per spec)	~60–66 °C (per spec)
Time	Per spec (often ~10–15 min)	Per spec
Hazard headline	Carcinogenic chromate + hot acid	Hot acid (no carcinogen)
Result	Activated, deoxidized, bondable surface	Same — chromate-free



HEADS-UP — CR VI IN THE FPL ETCH

If your shop runs the **dichromate FPL etch**, the bath and its mist contain **hexavalent chromium — a confirmed carcinogen and skin sensitizer**. Local exhaust, respiratory protection per your program, chromate-rated gloves, strict no-skin-contact, rigorous hygiene, and proper **chromate waste handling** are mandatory. Know whether your tank is FPL or P2 *before* you work it.



HEADS-UP — HOT STRONG ACID + HYDROGEN

Both FPL and P2 are **hot, strong acid** (burns, splash, mist) and etching aluminum releases **hydrogen** (flammable). Full acid PPE; ventilation on; ignition sources away; **add acid to water** for make-up.



HEADS-UP — DON'T ETCH THROUGH THE CLAD

On **Alclad** parts the pure-aluminum skin is thin. Over-etching can cut through it and expose the copper core — ruining corrosion protection *and* bond quality. Hold the **specified etch time**; longer is not better.



TECHNICAL STUFF — WHAT THE PRE-ETCH ACTUALLY DOES FOR THE BOND

The FPL/P2 etch doesn't just clean — it **removes the native oxide, strips smut, and chemically conditions the aluminum** so that the phosphoric anodize grows its open whisker structure uniformly. On the classic FPL chemistry, the etch even begins to develop a thin oxide microstructure of its own. The cleaner and more uniform the etched surface, the more consistent the anodic oxide — and consistency is bond reliability.

STEP-BY-STEP

1. Confirm which etch your tank is (**FPL/Cr VI or P2/chrome-free**) and its specific hazards/first-aid. 2. Verify concentration and temperature in range. 3. Immerse for the **specified time** (controls activation and metal/clad loss). 4. Agitate per procedure. 5. Withdraw, drain back, move **directly to rinse** — don't let etch dry on the part.

DO

- Know whether your tank is FPL (Cr VI) or P2.
- Hold the specified etch time exactly.
- Protect thin clad from over-etch.
- Move straight to rinse; full PPE incl. respiratory where required.

TOP MISTAKES

- Not knowing if the tank is chromate-bearing.
- Over-etching (clad breach, fatigue debit).
- Letting etch dry on the part before rinse.
- Skimping on chromate hygiene/waste handling.

SIGN-OFF — STATION 03

Teach-Back: What the etch does (deoxidize/activate for anodize + bond); the difference between FPL (Cr VI) and P2 (chrome-free) and which your tank is; why over-etching clad is dangerous; the chromate and hot-acid hazards and first response.

"I etched for the specified time at correct concentration/temperature, knew whether my tank was FPL (Cr VI) or P2, protected clad from over-etch, moved straight to rinse, and wore full PPE including respiratory protection where required."

Trainee: _____ Trainer: _____ Etch type (FPL/P2): _____

STATION 04 OF 12

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT — AND ON A BOND LINE, A CLEANLINESS GUARANTEE.”

PURPOSE & THE “WHY”

Wash the **etch film** off the part before it reaches the anodize tank. Carry etch drag-out forward and it **contaminates the phosphoric bath** and leaves residue on the future bond surface. The rinse knocks drag-out down by dilution. Cleanliness is tracked by **conductivity (µS/cm)** — *lower = cleaner*. **If the etch was a dichromate FPL, this rinse water carries chromate** — handle and dispose accordingly.



TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-current** cascade: parts move forward, water flows backward, so the cleanest water touches the part last. It saves enormous water — important on a line with this many rinses — and gives the cleanest possible handoff.

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating needed
Time	30–60 s with agitation	Longer for cavities/blind holes
Water source	DI preferred (esp. near anodize)	Impurities harm both bath and bond
Conductivity	< 500 µS/cm (target < 300)	Monitor at shift start
Flow	Continuous overflow / counter-current	Prevents stagnation



HEADS-UP

This rinse carries dragged-out acid (and **chromate** if FPL). Goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses**. Don’t let chromate-bearing rinse go untreated to drain.

DO

- Check conductivity at shift start; tell your lead if near/above limit.
- Transfer promptly from the etch — don’t let it dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting parts drip-dry between etch and rinse.
- Forgetting chromate handling on FPL rinse water.

SIGN-OFF — STATION 04

Teach-Back: Why carrying etch into the anodize tank causes problems; define drag-out/drag-in; what conductivity tells you and which way is “clean”; chromate handling if FPL.

“Rinse conductivity was within limit, flow was running, parts were rinsed the full time, and I handled any chromate-bearing water correctly with PPE.”

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm

STATION 05 OF 12

DEOXIDIZE / DESMUT

“ETCHING UNCOVERS THE ALLOY’S LEFTOVERS — ESPECIALLY COPPER. DEOX WASHES THEM AWAY.”

PURPOSE & THE “WHY”

On **copper-rich aerospace alloys (2024, 7075)** the etch can leave a dark **smut** (copper, plus silicon/iron) on the surface. Anodize or bond over smut and you get **dull, non-uniform oxide and weak bonds**. An **acid deoxidizer/desmut** removes it, leaving a bright, uniform, bondable surface. *(On some PAA flows the FPL/P2 etch already deoxidizes adequately and a separate deox is omitted — follow your spec. When used, deox chemistry depends on the alloy; some are fluoride-bearing.)*



REMEMBER

Smut after etch is normal on copper alloys. Deox’s whole job is to remove it. A part that goes into the anodize tank still gray/smitty comes out streaky and dull — and bonds poorly. Deox is the bright-surface gatekeeper.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Acid deox (sometimes fluoride-bearing or proprietary)	Match to the alloy
Concentration	Per product TDS	Titrate regularly
Temperature	Often ambient-warm	Per TDS
Time	~0.5–3 min	Until smut is gone, no longer
Result	Bright, uniform, active, bondable surface	No gray film remaining



HEADS-UP — STRONG ACID (AND POSSIBLY FLUORIDE)

Deox is acidic; **fluoride-bearing deox is especially hazardous** (hydrofluoric-type chemistry causes deep, delayed burns and is systemically toxic). Know exactly which chemistry your tank uses, read its SDS, wear full acid PPE, and know the **specific first-aid** (fluoride burns may require **calcium gluconate** per your plan). **Add acid to water** for any make-up.

STEP-BY-STEP

1. Confirm whether a deox step is required for this alloy/spec, and its chemistry/hazards. 2. Check concentration and temperature in range. 3. Immerse for the specified short time — just until smut clears. 4. Withdraw, drain, move to rinse. 5. Don’t over-dip — excessive deox can etch/pit some alloys or thin clad.

SIGN-OFF — STATION 05

Teach-Back: What smut is and where it came from (alloying elements, esp. copper, left by the etch); why anodizing/bonding over smut fails; that deox chemistry depends on the alloy; the special hazard/first-aid if fluoride-bearing.

“I removed the smut with the correct deox for the alloy, did not over-dip, and understand the specific hazard and first-aid for this tank (including fluoride if applicable).”

Trainee: _____ Trainer: _____ Chemistry: _____

STATION 06 OF 12

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE MAIN EVENT.”

PURPOSE & THE “WHY”

Remove deox acid before the part enters the anodize tank. Carrying deox drag-out into the phosphoric bath introduces unwanted ions (and, with a fluoride deox, **fluoride contamination** that can roughen the oxide). A clean rinse protects the anodize bath and the bonding surface.

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	30–60 s, agitation	Longer for cavities
Water	DI preferred	Cleaner handoff to anodize
Conductivity	Per shop spec	Lower = cleaner
Flow	Counter-current overflow	—



HEADS-UP

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the prior tank was a **fluoride** deox, this rinse carries fluoride — handle and dispose accordingly.



TIP

Move from this rinse into the anodize tank **promptly and consistently**. A long, variable dwell in air or rinse before anodizing can cause part-to-part variation in the oxide — and on a bonding line, **consistency is bond reliability**.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: Why a clean handoff to the anodize tank matters; what fluoride drag-in could do.

“The part was fully rinsed, conductivity acceptable, and I moved promptly and consistently to the anodize tank.”

Trainee: _____ Trainer: _____ Date: _____

STATION 07 OF 12 • THE MAIN EVENT

PAA ANODIZE — THE MAIN TANK

WHERE THE ALUMINUM’S OWN SURFACE BECOMES A THIN, OPEN, GLUE-ABLE OXIDE — AND WHERE THE PART IS THE ANODE

Take a breath. Up to now, every station has had one job: get the part clean, activated, and ready. **Station 07 is where the PAA oxide is actually created** — the tank that defines this whole line.

The big idea in one sentence: **we run the aluminum part as the *anode* in dilute phosphoric acid and use DC current (~10–15 V) to grow a thin, very open, whisker-like oxide out of the part’s own surface — an oxide built to be glued, then left unsealed.** The cast:

- **Anode** — *your aluminum part (positive)*. Where the oxide grows.
- **Cathode** — **stainless steel or lead** plates (negative). They carry current and bubble off hydrogen; they do **not** feed metal into the part.
- **Electrolyte** — **dilute phosphoric acid (H₃PO₄)**, ~10–12% (~100–120 g/L), at **room temperature (~20–25 °C / 68–77 °F)**, with agitation.

★

REMEMBER — THE ANODIZE TANK IS THE MIRROR IMAGE OF A PLATING TANK

In plating, the part is the **cathode** and metal lands on it. Here the part is the **anode** and **oxide grows out of it**. The cathodes don’t supply the coating — all of it comes from the part’s own aluminum plus oxygen from the bath. Forget this and the whole tank behaves like a mystery.

⚠

HEADS-UP — PHOSPHORIC ACID + DC + HYDROGEN

This tank holds **phosphoric acid** (corrosive — milder than sulfuric/chromic, but still real burns and mist), runs **DC current** (shock/arc at contacts), and bubbles **flammable hydrogen** off the cathodes. Engineering controls — **exhaust, agitation, splash control** — must be working before you run it. Full acid PPE. No ignition sources.

• WHY PHOSPHORIC (AND WHAT MAKES IT PARTICULAR)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Very open whisker oxide	The best mechanical anchor for adhesive/primer
Phosphate-rich, hydration-resistant skin	Durable bonds that resist humidity/bond-line corrosion
Thin, low-distortion	Right for tight-fit, fatigue-critical bonded structure
Voltage-controlled, self-limiting	Grows the right structure and stops — thin by design
Milder acid than sulfuric/chromic	Lower acute bath hazard than other anodize lines

...but the same chemistry demands care: it must hit the **right voltage** (structure depends on it), it’s run for a **short, controlled time**, it’s sensitive to **contamination** (which harms bondability), and — above all — the result must be **kept clean and never sealed**.

• THE BATH MAKE-UP — KNOW THE RANGES

**HEADS-UP**

These are **typical** PAA ranges for training context. **The controlling document is your customer/process spec — Boeing BAC5555, ASTM D3933, and your shop’s lab sheet.** Verify before you run; never anodize to a number from a training manual.

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Phosphoric acid (H ₃ PO ₄)	~10–12% (~100–120 g/L)	The electrolyte; grows + opens the oxide
Temperature	~20–25 °C (68–77 °F), room temp	Warmer/cooler shifts pore structure
Voltage (DC)	~10–15 V (“10-volt”/“15-volt” PAA)	The primary control — sets the open structure
Current density	Low; tapers as oxide forms	Voltage-controlled, not amp-hour-controlled
Time	~20–25 min	Short; structure-forming, not thickness-building
Oxide thickness	~0.1–1 μm (a few hundred nm)	Thin on purpose — a bond surface, not a coating
Cathodes	Stainless steel or lead	Carry current; bubble H ₂ ; don’t feed metal
Agitation	Mild (air / solution)	Even bath at the surface
Contamination (chloride, oils, drag-in)	Very low	Contamination ruins the bondable surface

**REMEMBER — THE NUMBERS TO CARRY**

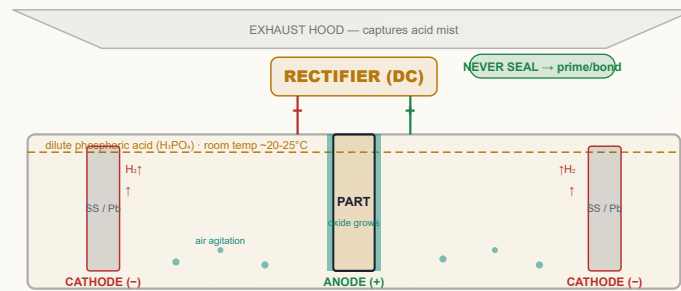
Phosphoric ~10–12%, room temp ~20–25 °C, ~10–15 V, ~20–25 min, thin oxide (~hundreds of nm), stainless/lead cathodes, mild agitation, **NEVER** sealed. These let you sanity-check almost anything about this tank.

**TECHNICAL STUFF — VOLTAGE BUILDS THE STRUCTURE**

In PAA you **ramp to the set voltage (e.g., 10 or 15 V) and hold it**. Higher voltage → a more open, larger-cell oxide; lower voltage → a tighter one. The classic “10-volt PAA” of ASTM D3933 and Boeing’s BAC5555 practice were chosen to give the **open whisker morphology that bonds best**. The current naturally **decays** as the thin insulating oxide forms — that decay is normal and is your sign the oxide reached its self-limited thickness.

**TECHNICAL STUFF — WHY THIN IS RIGHT**

Bond strength comes from the *interlock and chemistry at the surface*, not from oxide thickness. A thick oxide would add distortion and could be more brittle at the bond line. PAA deliberately stops at a thin, open, phosphate-topped oxide — the sweet spot for **durable** adhesion.



Key params: H₃PO₄ 10-12% • 20-25C • 10-15 V • ~20-25 min • thin open oxide • UNSEALED. Note the **reversed polarity** vs. a plating cell: the part is the ANODE (+).

• STANDARD OPERATING PROCEDURE

1. **Confirm the part is clean, etched, deoxed, rinsed, and racked tight** — straight from the rinse, no long air dwell.
2. **Verify engineering controls FIRST:** agitation on, exhaust on, splash control in place, bath at room-temperature range, full acid PPE on.
3. **Check bath readiness:** phosphoric in range (per lab), temperature in range, contamination within limits, cathodes present/sound.
4. **Confirm rectifier polarity: PART = ANODE (POSITIVE).** (*Opposite of plating — verify every time.*)
5. **Ramp to the target voltage (~10–15 V per spec)** and hold it.
6. **Hold for the specified time (~20–25 min)**, watching that the **current tapers** normally and temperature stays in range.
7. **At time, stop current, withdraw, drain over the tank**, and move to the post-anodize rinse — the pores are now **open and ready for prime/bond**.
8. **Do NOT route the part toward any seal tank.** The next steps are rinse → dry → prime/bond.
9. **Log everything:** voltage, time, temperature, current trend, checks.



HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME

Because everyone trained on plating expects the part to be the **cathode**, the most dangerous habit is assuming polarity. **The part is the ANODE (+).** Reversed polarity gives no coating (and can damage parts/cathodes). Check before you energize.

•

COMMON DEFECTS — WHAT YOU’LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Weak/failing bonds in test	Contaminated, sealed, or aged surface; wrong voltage/time	Verify no seal/contamination; check voltage/time; re-prep
Dull/gray, non-uniform oxide	Smut not removed; copper-rich alloy; deox issue	Confirm deox; confirm alloy; lab-check bath
Thin / no oxide	Bad contact; wrong polarity; low voltage	Re-rack tight; verify part = anode ; check voltage
Burned/rough at contact	Local current crowding; poor contact	Improve racking/contact
Pitting / roughness	Contamination (chloride, fluoride drag-in)	Lab-check bath; use DI; treat per chemist
Surface water-breaks after anodize	Oil/contamination, or accidental sealing	Re-clean/re-prep; eliminate the contamination source

•

INTEGRATED SAFETY — THIS STATION DEMANDS RESPECT

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HEADS-UP — PHOSPHORIC ACID (THE HEADLINE)
Corrosive (milder than sulfuric/chromic but still burns), splash and mist hazard. Engineering controls (exhaust, splash guards) first, PPE always. Any skin/eye contact → flush **15 minutes**, get medical attention. **Acid to water** for make-up.

⚠

HEADS-UP — HIGH-CURRENT DC & FLAMMABLE HYDROGEN
Shock/arc at contacts and bus bars — **LOTO** for all electrical work, dry hands only. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.

⚠

HEADS-UP — PROTECT THE BONDING SURFACE
The product of this tank is a *bond surface*. Contamination, sealing, or rough handling here can't be “fixed” later — it scraps the bond. Cleanliness discipline is a quality control here.

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge? (*Anode, +.*)
- Where does the coating come from? (*The part's own Al + O₂.*)
- The headline numbers (acid %, temp, voltage, time, thickness)?
- Why is PAA voltage-controlled and self-limiting?
- Why do we **never seal** a PAA part?
- The three safety families? (*Acid, DC, hydrogen — plus chromate/solvent elsewhere.*)

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it's the **anode**).
- Reaching for a seal tank out of habit.
- Loose rack contact (no-coat / burned contact).
- Bare-hand contact on bond surfaces.
- Long, variable dwell before anodizing.
- Eating/drinking/face-touching near the acid tank.

SIGN-OFF — STATION 07

“I can independently: verify agitation, exhaust, and PPE before running; verify bath readiness (acid, temp, contamination per lab); confirm the part is racked as the ANODE (positive); ramp to and hold the target voltage; control the oxide by voltage × time; recognize dull/thin/pitted defects; state the acid, electrical, and hydrogen hazards; and confirm the part goes to RINSE → DRY → PRIME/BOND and is NEVER sealed. I understand chemistry adjustments are made only by authorized personnel.”

Operator: _____ Date: _____ Trainer: _____ Date: _____

— STATION 08 OF 12

RINSE (POST-ANODIZE)

“GENTLE NOW — THE PORES ARE WIDE OPEN, AND THIS SURFACE IS ABOUT TO BE GLUED.”

PURPOSE & THE “WHY”

Rinse the phosphoric acid out of the freshly anodized part **without contaminating the open pores and without sealing them**, before drying and priming. The fresh oxide is **porous and absorbent** — it will soak up whatever it touches. Leftover acid interferes with priming/bonding. But the rinse must be **clean (DI)** and — importantly — **not hot enough to start sealing the oxide**.



REMEMBER — DON'T ACCIDENTALLY SEAL

Use **cool/ambient DI** here. **Hot water can begin to hydrate and close the open pores** — exactly what we must *not* do on PAA. Save no heat for “sealing,” because there is no sealing on this line. Cool, clean, gentle.

PARAMETER	RANGE	NOTES
Water	DI strongly preferred	Pores absorb impurities
Temperature	Cool / ambient	Avoid sealing the oxide
Time	1-3 min, gentle agitation	Flush acid from pores
Conductivity	Low (per spec)	Confirms acid removed



HEADS-UP

This rinse carries dragged-out **phosphoric acid** — mildly acidic, slip hazard, goggles/gloves. Don't let acid-laden rinse go untreated to drain (see Part Three / waste — phosphate is regulated).

STEP-BY-STEP

- Confirm cool DI flow.
- Rinse gently, full time.
- Drain over the tank.
- Move promptly to dry.

TOP MISTAKES

- **Hot water (seals the pores!)**
- Hard tap water (impurities).
- Short rinse leaving acid behind.
- Touching bond surfaces.

SIGN-OFF — STATION 08

Teach-Back: Why the pores are absorbent now; why cool DI (no hot water → no accidental sealing); why contamination here ruins the bond.

“I rinsed with cool DI to flush acid from the open pores without sealing or contaminating them, and moved promptly to dry.”

Trainee: _____ Trainer: _____ Date: _____

— STATION 09 OF 12 • REPLACES “DYE”

DRY (CLEAN & GENTLE)

“ON A COLOR LINE THIS SLOT IS THE DYE TANK. ON A PAA LINE, IT’S THE CAREFUL DRY THAT PROTECTS THE BOND.”

PURPOSE & THE “WHY”

Dry the rinsed part **completely, cleanly, and gently** so it’s ready for primer/adhesive. A wet or contaminated surface bonds poorly. But the dry must be **gentle and clean** — no oil-laden shop air, no excessive heat that could damage the thin oxide or, in the limit, begin hydrating/sealing it, and no airborne contamination settling on the bond surface.



REMEMBER — DRYING REPLACES DYEING HERE

A decorative line would dye now. PAA has **no dye and no seal**. The post-anodize sequence is **rinse → dry → prime/bond**. Drying is not a throwaway step — it delivers a clean, dry, open surface to the primer.

PARAMETER	RANGE	NOTES
Method	Clean warm forced air / low-temp oven	Use clean, oil-free air
Temperature	Warm, controlled (per spec)	Avoid excessive heat / oxide damage
Air quality	Filtered, oil-free	Compressed shop air often has oil — verify
Handling	By rack / clean gloves only	No bare-hand prints on bond surfaces



HEADS-UP

Parts and racks may be **warm** — burn/handling care. Drying ovens are a heat hazard; mind hot racks. **Oil-contaminated compressed air is a hidden bond-killer** — never blow off a bond surface with unverified shop air.



TIP — THE CLOCK STARTS HERE

Once the part is dry, the **bond/prime time window** is effectively running (Station 11 / Part Three). Keep the dried part **clean, covered, and moving toward primer** — don’t dry a part and let it sit out exposed.



TECHNICAL STUFF — WHY DRYING MATTERS MORE THAN IT LOOKS

Trapped water in the open pores, or oil deposited from dirty air, sits exactly where the primer must wet and anchor. A clean, complete, gentle dry hands the primer a *pristine, open, dry scaffold*. Anything left behind — moisture, oil, dust — becomes a weak spot in the cured bond.

SIGN-OFF — STATION 09

Teach-Back: Why this step replaces dye (PAA isn’t dyed); why clean/oil-free air and gentle heat matter; that the time window starts as the part dries.

“I dried the part completely with clean, oil-free air at controlled temperature, handled it only by rack/clean gloves, and kept it protected and moving toward priming.”

Trainee: _____ Trainer: _____ Date: _____

STATION 10 OF 12 • REPLACES “SEAL”

PRIME / BOND

“THE DECORATIVE LINE SEALS HERE. WE DO THE OPPOSITE — WE FILL THE OPEN PORES WITH PRIMER AND ADHESIVE.”

PURPOSE & THE “WHY”

Apply the **corrosion-inhibiting bond primer** and/or the **structural adhesive** to the freshly prepared, dried PAA surface so the polymer **wicks into the open whisker oxide and cures there**, creating a durable, interlocked bond. This is the payoff of the whole line — and it is the **exact opposite of sealing**.

★

REMEMBER — THE NO-SEAL RULE, STATED ONE MORE TIME

On Type II/BSAA, Station 10 *closes* the pores (seal). On PAA, Station 10 *fills* the pores with primer/adhesive. We never seal a PAA part. Sealing would shut the pores, block the primer, and destroy the bond. If you take one rule from this manual: **PAA is primed/bonded, not sealed.**

THE BOND PRIMER

A **corrosion-inhibiting adhesive primer** (a classic example is **BR 127–type primer** — *verify the exact primer and its spec/SDS for your job*) is usually applied to the PAA surface soon after anodizing. The primer:

- **Wets and penetrates** the open oxide, anchoring into the whiskers.
- **Protects the prepared surface** until the adhesive/assembly step.
- **Inhibits corrosion** at the bond line (many such primers are **chromated** — a Cr VI hazard to control).

PARAMETER	PRACTICE	NOTES
Primer type	Corrosion-inhibiting adhesive primer (per spec)	e.g., BR 127–type — verify spec/SDS
Application	Spray (or per spec), thin uniform film	Controlled thickness per TDS
Surface state	Unsealed, clean, dry PAA	Primed promptly within the time window
Cure	Per primer TDS (often a low-temp bake)	Don't exceed/undershoot the cure schedule
Hazards	Solvent, chromate, (some) isocyanate	Respiratory protection + ventilation

⚠

HEADS-UP — THE PRIME AREA IS A RESPIRATORY HAZARD ZONE

Bond primers carry **solvents, chromate pigments, and (in some products) isocyanates**. Use the **respiratory protection your program requires (often supplied-air for spray)**, proper booth ventilation, solvent-rated gloves, and eye protection. Chromate is carcinogenic/sensitizing; isocyanates can cause permanent respiratory sensitization. Read the primer SDS — this is not “just paint.”

**TECHNICAL STUFF — WHY PROMPT PRIMING MATTERS**

The fresh PAA oxide is open and active. Left exposed, it slowly **adsorbs contamination and hydrates** (the phosphate skin slows this, but doesn't stop it forever). Priming **promptly** locks in the prepared surface while it's at its best — which is why specs set a **time window** between anodizing and priming/bonding (Station 11 / Part Three).

**TIP**

Never touch a surface that's about to be primed, and never prime a surface you're not sure is clean and dry. If a dried part has been sitting, exposed, or possibly contaminated, **flag it** — re-prep is far cheaper than a failed bond on an aircraft.

STEP-BY-STEP

1. Confirm the surface is **unsealed, clean, dry PAA**, within the time window. 2. Confirm the correct primer and cure schedule (per spec/TDS). 3. In a properly ventilated area with required respiratory PPE, apply a **thin, uniform** primer film. 4. Cure per the primer TDS. 5. Route to adhesive/assembly or protect for transport. **At no point apply a seal.**

TEACH-BACK — CAN YOU ANSWER?

- Why we prime/bond instead of seal.
- That **PAA is never sealed**.
- How the primer anchors into the open oxide.
- The time-window reason for prompt priming.
- The chromate/solvent/isocyanate hazards and respiratory controls.

TOP MISTAKES

- Reaching for a seal tank out of habit (scraps the bond).
- Priming a contaminated or aged surface.
- Skipping respiratory protection in the prime area.
- Wrong/uneven primer film; bad cure schedule.
- Bare-hand contact on the surface before priming.

**REMEMBER**

This station is where PAA pays off — or fails. The primer/adhesive *fills* the open whisker oxide and cures into a 3-D interlock. Hand it a clean, dry, unsealed surface on time and you get a bond that lasts decades. Seal it, contaminate it, or wait too long, and you've scrapped everything the line built.

SIGN-OFF — STATION 10

"I confirmed an unsealed, clean, dry surface within the time window, applied the specified primer in a ventilated area with required respiratory PPE, cured it per TDS, and never applied any seal."

Trainee: _____ Trainer: _____ Primer/spec: _____

STATION 11 OF 12 • REPLACES “RINSE/DRY”

CURE / BOND ASSEMBLY

“A GREAT SURFACE PREP IS WASTED IF THE BOND IS MADE TOO LATE, TOO DIRTY, OR CURED WRONG.”

PURPOSE & THE “WHY”

Lay up and **cure the bonded assembly** (adhesive film, honeycomb core, doublers, laminate) on the primed PAA surface, **within the allowed time window** and to the correct cure schedule. PAA’s durability advantage is only realized if the bond is actually made on a fresh, clean, properly primed surface and cured correctly.

**REMEMBER — THE TIME WINDOW IS REAL**

Specs set a **maximum time** from anodizing to priming, and from priming to bonding/assembly. Miss the window, or let the surface get dirty or humid, and the bond loses the very durability PAA exists to provide. **Know your window and respect it.** (*Windows vary by spec — confirm against your customer process spec; keep parts clean, dry, and covered while they wait.*)

PARAMETER	PRACTICE	NOTES
Time window	Per spec (anodize→prime, prime→bond)	Don’t exceed; keep surface clean/dry/covered
Adhesive	Per spec (film/paste)	Correct lay-up, no contamination
Cure	Per adhesive spec (heat/pressure, often autoclave/press)	Temperature, pressure, time all matter
Cleanliness	Controlled — clean room / clean bench	Dust, oils, fingerprints all weaken bonds

**HEADS-UP — CURE EQUIPMENT HAZARDS**

Bond cure often uses **heat and pressure (ovens, presses, autoclaves)** — burn, pinch, and pressure hazards. Follow the equipment procedures and your shop’s bonding controls.

**TIP**

Treat every minute the prepared surface is exposed as a small risk to the bond. The disciplined bonder **moves parts steadily from anodize → dry → prime → bond** and never lets a prepared surface “wait around.”

SIGN-OFF — STATION 11

Teach-Back: What the time window is and why it matters; why cleanliness and correct cure are bond-critical; the cure-equipment hazards.

“I made/assembled the bond within the time window on a clean, properly primed surface, cured per spec, and respected the cure-equipment hazards.”

Trainee: _____ Trainer: _____ Date: _____

STATION 12 OF 12

FINAL INSPECTION

“PROVE THE PREP — AND PROVE THE BOND. ON PAA, THE WEDGE TEST IS KING.”

PURPOSE & THE “WHY”

Verify the surface prep and the **bond quality**. PAA quality isn’t visible by eye. **Surface-prep verification** (water-break, and per spec) confirms a clean, properly prepared surface; **process-control coupons** anodized and bonded alongside the parts are pulled and tested — most famously with the **wedge crack test (ASTM D3762)** — to prove the bond is **durable**, not just strong on day one.

CHECK	METHOD (TYPICAL STD)	ANSWERS
Surface cleanliness	Water-break test; per spec	“Clean, bondable surface?”
Surface prep verification	Visual, water-break, and per spec (surface analysis on coupons)	“Did PAA prep correctly?”
Oxide / morphology (coupon)	Per spec (microscopy where required)	“Right open oxide grown?”
Bond durability	Wedge crack test — ASTM D3762 (coupons)	“Will the bond <i>last</i> in humidity?”
Bond strength	Lap-shear / peel per spec (coupons)	“Strong enough?”
Appearance	Visual under good light	No contamination, damage, or (forbidden) sealing



REMEMBER — THE TEST THAT MATTERS MOST ON PAA

The **wedge crack test (ASTM D3762)** is the signature PAA quality check: a wedge is forced into a bonded coupon and the **crack growth is measured after humidity exposure**. A good PAA bond resists crack growth and fails **cohesively** (in the adhesive), not at the metal interface. **Durability — not just initial strength — is the whole point of PAA**, and the wedge test is how we prove it.



TIP — THE WEDGE TEST IN PLAIN TERMS

Drive a wedge into the bonded coupon, mark the crack tip, expose it to heat/humidity, and re-measure. **Little crack growth + cohesive failure = a durable PAA bond**. Lots of growth or **interfacial failure (clean metal showing) = a bad prep** — re-examine the whole line (contamination? sealing? bad etch? missed time window?).

STEP-BY-STEP

- Water-break check; verify surface prep.
- Pull and test process-control coupons (wedge ASTM D3762, lap-shear/peel per spec).
- Visual under good light.
- Accept / reject / route to re-prep.

TOP MISTAKES

- Skipping coupons.
- Judging a bond by eye.
- Ignoring interfacial (metal-side) failures.
- Missing evidence of contamination or accidental sealing.

SIGN-OFF — STATION 12

Teach-Back: Which check proves *durability* (wedge crack ASTM D3762) vs. strength; what cohesive vs. interfacial failure means; why surface-prep verification matters before bonding.

“I verified surface cleanliness/prep, ran the process-control coupons including the wedge crack test, interpreted cohesive vs. interfacial failure, and accepted/rejected accordingly.”

Trainee: _____ Trainer: _____ Wedge result / failure mode: _____

PART THREE — ACROSS THE LINE
THE CROSS-CUTTING TOPICS

ACROSS THE LINE

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

• BOND DURABILITY & THE WHISKER MORPHOLOGY (DEEP DIVE)

PAA's reputation rests on one thing: **durable bonds**. Two features deliver it. First, the **open, whisker-like oxide** gives the adhesive/primer a **3-D mechanical interlock** — cured polymer is locked into a microscopic scaffold, not just stuck to a smooth face. Second, the **phosphate-rich oxide resists hydration** — it slows the moisture attack that, on weaker preps, creeps along the bond line and peels joints apart over years. Together they make bonds that pass the **wedge crack test (ASTM D3762)** by resisting crack growth and failing **cohesively**. *Thickness is not the point — the open structure and hydration resistance are.*



REMEMBER

PAA = **open whiskers (interlock) + phosphate skin (hydration resistance) = durable bonds**. Everything on the line exists to deliver that surface clean, undisturbed, and on time.

• THE NO-SEAL RULE AND WHY (DEEP DIVE)

Sealing closes pores. On a decorative/protective line that's the goal (corrosion resistance, locked-in dye). On **PAA it is forbidden** — closed pores can't be penetrated by primer/adhesive, so a sealed PAA part has **no interlock and a weak, undurable bond**. PAA gets its durability from **chemistry (phosphate) + the cured bond filling the pores**, not from sealing. So: **never seal; keep the post-anodize rinse cool; avoid any hot step that could hydrate the oxide; dry clean; prime/bond promptly.**

• THE PRIME/BOND TIME WINDOW (DEEP DIVE)

A fresh PAA surface is at its best right after preparation. Left exposed it slowly **adsorbs contamination and hydrates**. Specs therefore set a **time window** — a maximum time from anodize to prime, and from prime to bond. Within the window the surface gives its full durability; past it, durability drops. (*Exact windows are spec-dependent — confirm against the customer process spec. Keep waiting parts clean, dry, and covered.*) The disciplined shop moves parts **steadily** through anodize → dry → prime → bond.



REMEMBER

Three PAA-only rules govern the back half of the line: **(1) never seal, (2) keep it clean** (no fingerprints/oil/dust on bond surfaces), **(3) prime/bond within the time window**. Break any one and the bond suffers.

• **RACKING & ELECTRICAL CONTACT (DEEP DIVE)**

The part is the anode, so **the rack is the wire that grows your oxide**. Good racking means: firm, current-carrying contact; **contact placed off any bond surface** where the bare mark won't matter; orientation for gas escape and drainage (no trapped air = no un-anodized voids on a bond face); no parts touching (shadowing); and maintained racks (titanium preferred; aluminum racks must be stripped/re-pointed as they anodize). On PAA, racking also protects the **cleanliness** of the bond surfaces. *(Cross-ref Station 01.)*

• **ALUMINUM ALLOY EFFECTS (DEEP DIVE)**

On bonding alloys, two facts dominate: **copper (2024/7075) makes heavy smut** — so deox discipline is mandatory — and **clad (Alclad) skins are thin** — so over-etching is dangerous. Rack like-alloy with like-alloy, and confirm clad-vs-bare before setting etch time. *(Cross-ref Chapter 6.)*

• **RINSING & WATER/DI MANAGEMENT**

Rinses are firewalls between chemistries *and* cleanliness guarantees for the bond. Use **counter-current cascade** rinsing; track **conductivity** (lower = cleaner); use **DI water** near anodize and post-anodize; and keep the **post-anodize rinse cool** so it never starts sealing the oxide. Tap-water impurities cause contamination that weakens bonds.



TECHNICAL STUFF — WHY COUPONS MATTER SO MUCH ON PAA

A PAA oxide is a few hundred nanometers thick — far too thin to judge by eye or routine gauge, and destructive checks would scrap a flight part. So shops run **process-control coupons** of the same alloy through the same load, then test *those* for surface prep, morphology, lap-shear/peel, and the **wedge crack test** — proving the process without harming the hardware.



REMEMBER

Across the whole line, the PAA operator's job is constant: **deliver a clean, open, undisturbed oxide to the primer, on time, never sealed**. Racking, alloy care, and rinsing all serve that one goal.

CHROMATE-FREE CHOICES & ENVIRONMENTAL

CR VI NOTE & ENVIRONMENTAL

WHERE THE CHROMIUM HIDES ON A PAA LINE — AND HOW THE WASTE STREAMS ARE HANDLED

• FPL CR VI NOTE & CHROMATE-FREE ALTERNATIVES

The classic FPL etch is sulfuric–sodium-dichromate — hexavalent chromium (Cr VI), a carcinogen. It performs well but carries the chromium hazard and chromate waste burden. The chromate-free P2 etch (ferric-sulfate/sulfuric) was developed as a drop-in replacement and is widely used to get Cr VI off the line. Likewise, many bond primers are chromated, and chromate-free primers exist. Where your spec allows, chrome-free options (P2 etch, chrome-free primer) reduce the worst hazard on the line. *(Always confirm what your customer spec permits — substitution must be qualified, not assumed.)*

**HEADS-UP**

Know whether your **etch (FPL vs P2)** and your **primer (chromated vs chrome-free)** contain Cr VI. That single fact drives your respiratory protection, skin protection, hygiene, and waste handling.

• WASTE & ENVIRONMENTAL

A PAA line generates several regulated streams — **never to a floor drain untreated**:

- **Spent phosphoric acid & acid rinses** — neutralize/treat per permit; **phosphate (nutrient) discharge is regulated** — manage it.
- **FPL etch waste (if used)** — **acidic + hexavalent chromium**: requires **Cr VI reduction/treatment** and chromate waste handling. (P2 etch waste is chromate-free but acidic + iron — still treated.)
- **Caustic cleaner waste** — high pH; neutralize/treat.
- **Deox/desmut waste** — acidic; **fluoride-bearing** deox needs fluoride treatment (precipitate as calcium fluoride).
- **Primer/solvent & overspray waste** — **solvent, chromate, isocyanate**: hazardous waste; capture overspray; manage VOCs and chromate per permit.
- **Air** — acid mist, **chromate mist (FPL/primer)**, and solvent VOCs need exhaust/scrubbing per permit.

**HEADS-UP**

Acidic and caustic streams must be **neutralized under control** (never casually mixed); **chromate and fluoride streams need dedicated treatment**. The wrong drum or drain can be a reportable violation. Know your shop's segregation scheme.

**FUN FACT**

Many bonding shops have moved from the dichromate FPL etch to the chromate-free **P2 etch** specifically to get Cr VI off the line — the same motivation that created BSAA to replace chromic anodizing. The industry keeps finding ways to keep the performance and drop the carcinogen.

— HOW WE PROVE IT

QUALITY CHECKS & TEST METHODS

THE CHECKS THAT CONFIRM A PAA PART IS RIGHT — CLEAN SURFACE, AND (ABOVE ALL) A DURABLE BOND

CHECK	METHOD (TYPICAL STD)	ANSWERS
Surface cleanliness	Water-break test	"Clean enough to anodize/bond?"
Surface prep verification	Visual + per spec (surface analysis on coupons)	"Did PAA prep correctly?"
Oxide morphology	Microscopy on coupons (per spec)	"Right open whisker oxide?"
Bond durability	Wedge crack test — ASTM D3762	"Will the bond <i>last</i> (humidity)?"
Bond strength	Lap-shear / peel per spec	"Strong enough now?"
Failure mode	Examine fracture (cohesive vs interfacial)	"Good prep (cohesive) or bad (interfacial)?"
Appearance/defects	Visual under controlled light	Contamination, damage, evidence of sealing



REMEMBER

Water-break = "clean surface?", wedge crack test (ASTM D3762) = "durable bond?", and failure mode (cohesive vs interfacial) = "good prep or bad prep?" Those three answer the questions that matter on a PAA line.



TIP — READ THE FRACTURE, NOT JUST THE NUMBER

A bonded coupon can show decent strength on day one and still be a *bad* PAA part. The honest tell is the **failure mode**: a bond that tears *through the adhesive* (cohesive) proves the prep gripped; a bond that peels off *clean metal* (interfacial) proves the prep failed — investigate contamination, sealing, etch, or a missed time window.



TECHNICAL STUFF — WHY THE WEDGE TEST, SPECIFICALLY

The wedge crack test (ASTM D3762) deliberately stresses the *weakest* thing about a bond: its resistance to **moisture-driven crack growth at the interface**. By wedging a coupon open and then soaking it in heat/humidity, it exposes a poor prep in hours that service might take years to reveal. That's why it's the signature PAA durability check — it tests for the exact failure PAA exists to prevent.



FUN FACT

The wedge test is almost theatrical: a metal wedge is literally hammered into the bonded coupon, the crack tip is marked, and the coupon goes into a hot, humid chamber. Engineers come back hours later and measure how far the crack crept. A great PAA bond barely moves.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Bond fails wedge test / interfacial failure	Contaminated, sealed, aged, or poorly etched surface	Re-examine clean/etch/deox; confirm no sealing ; check time window; re-prep
Weak bond strength	Oil/fingerprints on bond surface; bad prep	Re-clean; protect bond surfaces; verify water-break
Dull / gray / non-uniform oxide	Smut not removed; copper alloy; deox issue	Verify deox; confirm alloy; lab-check bath
Thin / no oxide	Poor contact; wrong polarity (part not anode) ; low voltage	Re-rack tight; verify part = anode ; check voltage
Pitting / roughness	Contamination (chloride/fluoride); over-etch	DI water; lab-check bath; control etch/deox time
Etched through clad	Over-etch on thin Alclad	Reduce etch time; protect clad
Surface water-breaks after prep	Oil/contamination, or accidental sealing	Eliminate contamination; no hot rinse ; re-prep
Bond made too late	Time window exceeded; surface aged	Respect window; keep parts clean/dry/covered; re-prep
Primer adhesion poor	Sealed/contaminated/aged surface; wrong cure	Confirm unsealed/clean; prime promptly; cure per TDS



TIP

Most PAA failures trace to one of four roots: **contamination** (oil/fingerprints/dust), **the no-seal rule broken** (accidental sealing/hot water), **bad pre-etch/deox** (smut, over-etch, wrong chemistry), or **the time window missed**. Name the root, not just the symptom.



REMEMBER

Two roots are also *safety-relevant*: **wrong polarity** (verify part = anode) and **chromate handling** (FPL etch / chromated primer). When in doubt about a change, ask your supervisor before you act.

— EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

Adhesive bonding: joining parts with a structural adhesive instead of (or with) fasteners. PAA prepares aluminum for it.

Anode: the **positive** electrode. In anodizing it's **your part** — where the oxide grows.

Anodizing: growing an integral oxide layer out of a metal's surface by making the part the anode in an electrolyte.

BAC5555: Boeing's process specification for **phosphoric acid anodizing for adhesive bonding**.

Barrier layer: the thin, dense oxide at the bottom of the coating, against the metal.

Bond primer (corrosion-inhibiting adhesive primer): a thin primer (e.g., BR 127-type) applied to PAA soon after anodizing to anchor into the oxide, protect the surface, and inhibit bond-line corrosion. Often chromated.

Cathode: the **negative** electrode (stainless/lead here) — carries current and bubbles off hydrogen; does **not** feed the coating.

Clad (Alclad): a thin pure-aluminum skin roll-bonded onto a stronger core; anodizes cleanly but is thin (don't over-etch).

Cohesive vs interfacial failure: a bond that breaks *within the adhesive* (cohesive = good prep) vs. *at the metal/oxide* (interfacial = bad prep).

Deoxidize / desmut: an acid dip removing the dark **smut** (esp. copper) left by etching.

Dimensional growth: the coating grows ~half in, half out — tiny on thin PAA.

FPL etch (Forest Products Laboratory): the classic **sulfuric–sodium-dichromate** pre-etch — contains **Cr VI**.

P2 etch: the **chromate-free** (ferric-sulfate/sulfuric) replacement for FPL.

PAA: Phosphoric Acid Anodizing — the thin, open, **unsealed** anodize used as **surface prep for structural adhesive bonding**.

Phosphate layer: the thin phosphate-rich skin PAA leaves; resists hydration → durable bonds.

Porous / whisker layer: the open upper oxide PAA grows — the scaffold the adhesive locks into.

Sealing: closing the pores (hot water / nickel salts). **Done on decorative/protective anodize — NEVER on PAA.**

Time window: the spec-set maximum time from anodize→prime→bond before the surface degrades.

Valve metals: metals that anodize to form protective oxides (aluminum, titanium, magnesium, tantalum, niobium).

Water-break test: a free cleanliness check — clean metal sheets water; dirty metal beads it.

Wedge crack test (ASTM D3762): the signature PAA durability test — measures crack growth in a wedged bonded coupon after humidity exposure.



REMEMBER

Five terms matter most: **anode** (your part), **open/whisker oxide** (the bond scaffold), **never seal / prime instead** (the defining rule), **phosphate layer** (hydration-resistant durability), and **wedge crack test** (how we prove durability). Master those and you understand PAA.



FUN FACT

The word “**anode**” comes from the Greek *ánodos*, “a way up” (Faraday's term for the electrode where current enters the cell). On this line, that “way up” is literally where your bonding surface is born.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE / SDS / hygiene — phosphoric acid, FPL Cr VI/chromate, caustic, fluoride, hydrogen, DC, primer solvents/isocyanate (all stations)	—	—	—	—
2	Station 01 — Inspection & Racking (contact + bond-surface protection)	—	—	—	—
3	Station 02 — Clean / Degrease + water-break	—	—	—	—
4	Station 03 — FPL / P2 Etch (know which; Cr VI discipline; clad care)	—	—	—	—
5	Station 04 — Rinse + conductivity (chromate handling)	—	—	—	—
6	Station 05 — Deoxidize / Desmut (alloy-dependent)	—	—	—	—
7	Station 06 — Rinse	—	—	—	—
8	Station 07 — PAA Anodize (part = anode; phosphoric; ~10–15 V; never seal)	—	—	—	—
9	Voltage/time control & (tiny) dimensional growth	—	—	—	—
10	Station 08 — Post-anodize rinse (cool DI, open pores, no sealing)	—	—	—	—
11	Station 09 — Dry (clean, oil-free; time window starts)	—	—	—	—
12	Station 10 — Prime / Bond (no seal ; primer hazards)	—	—	—	—
13	Station 11 — Cure / assembly & the time window	—	—	—	—
14	Station 12 — Final Inspection & bond quality (wedge ASTM D3762)	—	—	—	—
15	LOTO on rectifier / electrical safety	—	—	—	—
16	Emergency response — eyewash, shower, spill, fluoride/chromate first-aid, solvent/isocyanate	—	—	—	—
17	Waste segregation & environmental (phosphate, Cr VI, fluoride, solvent/primer)	—	—	—	—
18	Troubleshooting — symptom recognition (incl. bond failure modes)	—	—	—	—

Operator name: _____ Hire/start date: _____

Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



HEADS-UP

Safety rows (1, 15, 16, 17) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS/hygiene (including **chromate**, **fluoride**, and **primer solvent/isocyanate** controls), LOTO, emergency response, and waste-control awareness are signed. Safety competency precedes production competency.

PART FOUR — CERTIFICATION

20 QUESTIONS • PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

**REMEMBER**

Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q3, Q9, Q13, Q17, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs: 3, 9, 13, 17, 19**

Q1. Anodizing differs from electroplating mainly because in anodizing the part is the:

- A. Cathode, and a different metal is deposited on it B. Anode, and an oxide is grown out of the part itself C. Cathode, and it dissolves D. Insulator, and nothing happens

Q2. The PAA anodize electrolyte is:

- A. Chromic acid (hexavalent chromium) B. Sodium hydroxide C. Dilute phosphoric acid (~10–12%) D. Concentrated sulfuric acid only

Q3. [SAFETY GATE] Which statement about PAA safety is *correct*?

- A. Phosphoric acid is milder than other anodize acids, but the chain still has hazards — a dichromate FPL etch contains Cr VI, plus caustic cleaner, fluoride deox, hydrogen, DC, and primer solvents/chromate B. PAA is completely hazard-free C. The phosphoric anodize bath contains the worst carcinogen on the line D. There is no chromium anywhere on a PAA line, ever

Q4. Typical PAA coating thickness is about:

- A. 25–100 µm (hardcoat range) B. A few hundred nm up to ~1 µm (very thin) C. 0.5–1 mm D. There is no coating

Q5. On a PAA line, the porous oxide is used mainly to:

- A. Grip primer and adhesive for durable structural bonds (a bond surface) B. Hold deep decorative dye color C. Make the part electrically conductive D. Add significant thickness

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is and **where the coating comes from**.

Q7. “Dimensional growth” in anodizing means the coating grows roughly:

- A. All outward B. All inward C. Half into the surface and half out (tiny on thin PAA) D. It shrinks the part

Q8. For a typical PAA aerospace part that will be **structurally bonded**, the usual practice is:

- A. Dye it a bright color, then seal heavily B. Apply Type III hardcoat first C. Seal the pores shut for corrosion resistance D. Leave it unsealed, dry it cleanly, and prime/bond promptly within the time window

Q9. (Short answer) [SAFETY GATE] Name the **two corrosive hazards** on the line (one high-pH, one low-pH), and state the universal rule for adding acid during bath make-up.

Q10. Typical PAA anodize voltage and temperature are about:

- A. 200 V; 0 °C B. ~10–15 V; ~20–25 °C (room temp) C. 40 V; held cold near 0 °C D. 1 V; 90 °C

Q11. A 2024 or 7075 part comes out **dull/gray** even though the line is running fine. The most likely cause is:

- A. The bath is too clean B. Too much dye C. The rectifier is off D. A copper-rich aerospace alloy (copper makes heavy smut) and/or smut not fully removed at deox

Q12. (Short answer) What is **smut**, where does it come from, and which station removes it? (Note why it's heavier on aerospace bonding alloys.)

Q13. [SAFETY GATE] The classic **FPL pre-etch** is best described as:

A. A sulfuric-acid + sodium-dichromate etch that contains hexavalent chromium (Cr VI) — a carcinogen; the chromate-free P2 etch is the substitute B. A harmless soap solution C. The same chemistry as the phosphoric anodize bath D. A nickel-sealing bath

Q14. Why was PAA developed (Boeing BAC5555)?

A. To make a thicker, harder coating than hardcoat B. To prepare aluminum surfaces for durable structural adhesive bonding (humidity/bond-line-corrosion resistant) C. To allow brighter decorative dye colors D. To run at much higher voltage than chromic

Q15. PAA oxide formation is controlled mainly by:

A. Dye color B. Seal temperature C. The alloy only D. The anodizing voltage and time (it is a voltage-controlled, self-limiting process)

Q16. (Short answer) State the “**no-seal**” rule for PAA and explain **why** sealing must not be done (what would happen to the bond).

Q17. (Short answer) [SAFETY GATE] List **three** pieces of PPE required on the PAA line, and name the special **respiratory** concern in the **prime/bond** area.

Q18. The PAA process is described/specified by:

A. Only MIL-PRF-8625 Type II B. Only the architectural anodize standard C. Boeing BAC5555 and ASTM D3933 (phosphoric acid anodizing for structural adhesive bonding) D. There is no standard for PAA

Q19. (Short answer) [SAFETY GATE] Name the **flammable gas** produced at the cathode during anodizing and **two** controls that keep it from becoming a fire hazard.

Q20. (Short answer) Give **two** reasons a PAA-prepared surface produces such **durable** adhesive bonds.



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 3, 9, 13, 17, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD



HEADS-UP

Keep this page out of the trainee's copy. Questions **3, 9, 13, 17, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Q1. B — The part is the **anode**, and oxide is grown out of the part itself.

Q2. C — **Dilute phosphoric acid (~10–12%, ~100–120 g/L).**

Q3. A — Phosphoric acid is **milder** than other anodize acids, but the chain still has hazards: **a dichromate FPL etch contains Cr VI**, plus caustic cleaner, fluoride deox, hydrogen, DC, and primer solvents/chromate. Safer in the anodize tank ≠ hazard-free.

Q4. B — **A few hundred nm up to ~1 µm** — thin by design (a bond surface, not a coating).

Q5. A — The open pores **grip primer/adhesive** for a durable structural bond — PAA is a bond surface, not decorative.

Q6. In plating the part is the **cathode** and a **different metal is deposited onto it** from the bath. In anodizing the part is the **anode**, and the coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath + the part's aluminum) — nothing is deposited from elsewhere.

Q7. C — Half in, half out; each surface grows out ~half the thickness (essentially negligible on thin PAA).

Q8. D — **Leave it unsealed**, dry cleanly, and **prime/bond promptly within the time window**.

Q9. The **caustic cleaner** (high pH / sodium hydroxide) and the **acid etch/anodize** (low pH — phosphoric, and sulfuric in the FPL/P2 etch) — both cause severe burns. Universal rule: **always add ACID to WATER, never water to acid**.

Q10. B — **~10–15 V; ~20–25 °C** (low voltage, room temperature — not refrigerated like a cold sulfuric tank).

Q11. D — A **copper-rich alloy (2024/7075)** makes heavy smut and dulls the oxide, and/or **smut wasn't fully removed** at deox.

Q12. **Smut** is the dark film of alloying elements — especially **copper** on aerospace alloys, plus silicon/iron — **left behind by the etch** when aluminum dissolves. Removed at the **deoxidize/desmut** station (acid dip). Heavier on **copper-rich 2xxx/7xxx** bonding alloys.

Q13. A — The classic **FPL etch is sulfuric acid + sodium dichromate — hexavalent chromium (Cr VI), a carcinogen**. The chromate-free **P2 etch** is the substitute.

Q14. B — To **prepare aluminum for durable structural adhesive bonding** (resisting humidity and bond-line corrosion) — Boeing BAC5555.

Q15. D — The **anodizing voltage and time** — PAA is a **voltage-controlled, self-limiting** process (ramp to ~10–15 V and hold).

Q16. No-seal rule: a PAA part is **never sealed**. Sealing closes the pores; the primer/adhesive could no longer penetrate the open whisker oxide, so there'd be **no mechanical interlock and a weak, undurable bond**. Instead of sealing, PAA parts are **dried and primed/bonded promptly**.

Q17. PPE (any three): sealed chemical splash goggles, face shield, chemical-resistant gloves, apron/suit, chemical-resistant boots. **Prime-area respiratory concern:** bond primers contain **solvents, chromate, and (some) isocyanates** — requires **respiratory protection (often supplied-air for spray) and proper ventilation**; chromate is carcinogenic and isocyanates can cause permanent respiratory sensitization.

Q18. C — **Boeing BAC5555 and ASTM D3933** (phosphoric acid anodizing for structural adhesive bonding).

Q19. Hydrogen gas (at the cathode). Controls (any two): **ventilation/exhaust running, no open flames/sparks/grinding nearby**, no smoking, keep ignition sources away.

Q20. Any two of: the **open, whisker-like oxide** gives the adhesive a **3-D mechanical interlock**; the **phosphate-rich oxide resists hydration** (moisture/bond-line corrosion); the surface is **kept unsealed** so primer/adhesive penetrates; the part is **primed/bonded promptly** within the time window on a clean, dry surface.

Answer-key letter distribution (MC): A: Q3, Q5, Q13 · B: Q1, Q4, Q10, Q14 · C: Q2, Q7, Q18 · D: Q8, Q11, Q15 — a healthy spread across all four letters (3 / 4 / 3 / 3 across the 13 multiple-choice items; the other seven are short-answer).



REMEMBER

A score of 16/20 passes — but every safety question (3, 9, 13, 17, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people.

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CERTIFICATE OF
COMPLETION

PHOSPHORIC ACID ANODIZING (PAA) TRAINING
PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Phosphoric Acid Anodizing (PAA)** training course — covering the full bonding-prep workflow (Inspection & Racking, Clean, FPL/P2 Etch, Rinse, Deoxidize, PAA Anodize, Rinse, Dry, **Prime/Bond — no seal**, Cure/Assembly, and Final Inspection), the fundamental concept that **the part is the anode and the oxide is grown out of the part itself**, the **open whisker-oxide / never-seal / prime-instead** principle, the prime-and-bond **time window**, oxide formation by voltage and time, electrical-contact and racking practice, aluminum-alloy and clad effects, bond-durability and the **wedge crack test (ASTM D3762)**, and — above all — the **safety practices** required of every operator (phosphoric-acid burn control, **Cr VI control in the FPL etch and chromated primers**, caustic and fluoride control, hydrogen and DC-electrical controls, **primer solvent/isocyanate respiratory controls**, spill response, and waste segregation) — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/aerospace specifications (e.g., Boeing BAC5555, ASTM D3933, and applicable bond-primer/process specs), and applicable regulatory requirements before production use. Phosphoric acid causes burns; a dichromate FPL etch and many bond primers contain hexavalent chromium (Cr VI); primers may contain isocyanates — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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